Vectors

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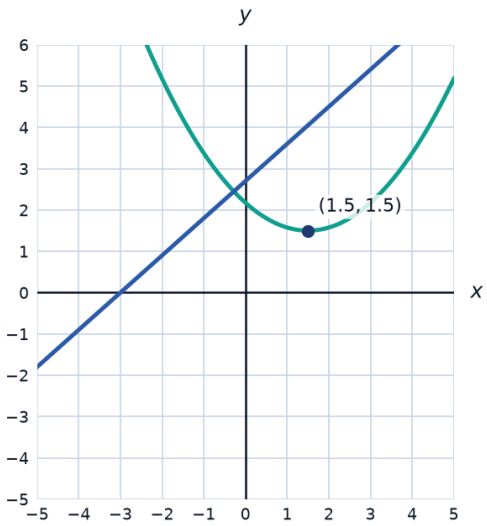
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.

2 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.



3 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.

4 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.

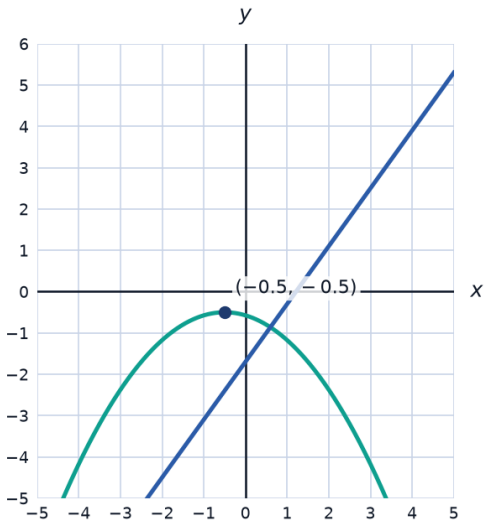
5 **[Foundational]** • **Target: The vector equation of a line**

Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.

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6 [Foundational] • Target: The vector equation of a line

Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.



7 [Foundational] • Target: The vector equation of a line

Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.

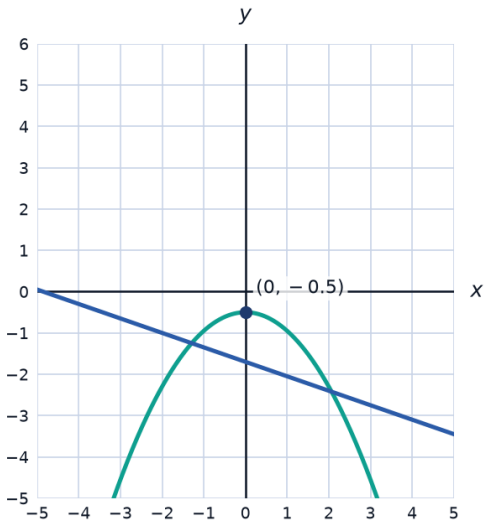
9 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.

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10 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.



11 [Intermediate] • Target: Intersecting and skew lines

Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.

12 [Intermediate] • Target: Intersecting and skew lines

Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.

13 [Intermediate] • Target: The equation of a plane

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

14 [Intermediate] • Target: The equation of a plane

Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

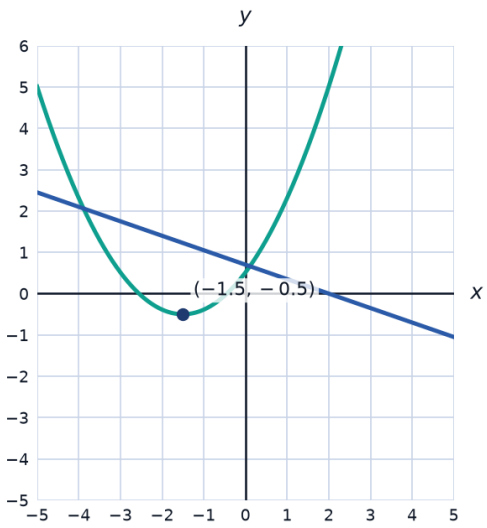
15 [Challenging] • Target: The equation of a plane • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.

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16 [Challenging] • Target: The equation of a plane • Cross-Lesson: 9.01 Complex Numbers

Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.



17 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.

18 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 10.01 Numerical Methods

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

19 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 11.01 Differential Equations

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

20 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 12.01 Poisson Distribution

Find the foot of the perpendicular from $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$, and the shortest distance from P to the line.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vector arithmetic in three dimensions

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. x = 60 m, y = 30 m

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

22 [Foundational] • Target: The scalar (dot) product and the angle between vectors

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$

b. $\langle -3, -4 \rangle$

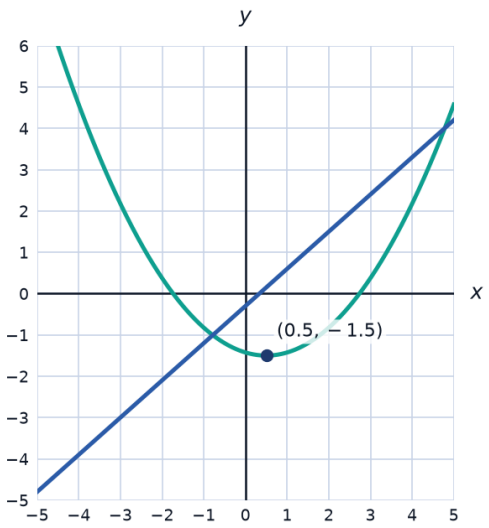
c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

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23 [Foundational] • Target: The vector equation of a line

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:



a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

24 [Foundational] • Target: Classifying lines: parallel, perpendicular and collinear points

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. 20

c. 25

d. 22

25 [Foundational] • Target: Intersecting and skew lines

A line has direction $\mathbf{d} = (2;\ -1; 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0°

d. 90°

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26 [Foundational] • Target: The equation of a plane

Dot product $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$. For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} =$

a. 1

b. -1

c. 0

d. $|a||b|$

27 [Foundational] • Target: Angles between lines, planes, and pairs of planes

Find the angle between planes $x + y = 0$ and $x + z = 0$:

a. $\pi/3$

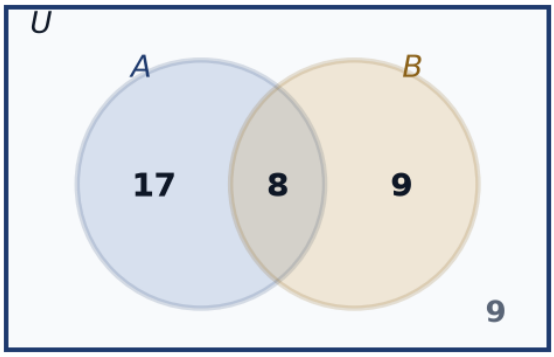
b. 0

c. $\pi/2$

d. $\pi/4$

28 [Foundational] • Target: Distances and same-side reasoning

Line of intersection of two non-parallel planes: direction vector $=$



a. sum of normals

b. cross product of normals

c. difference of normals

d. dot product of normals

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a drone's flight path

Two vectors are anti-parallel if angle between them is:

a. 0

b. $\pi/4$

c. π

d. $\pi/2$

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30 [Intermediate] • Target: Vector arithmetic in three dimensions

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

31 [Intermediate] • Target: The scalar (dot) product and the angle between vectors

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. (2, -3)

d. (-3, -2)

32 [Intermediate] • Target: The vector equation of a line

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. -6

b. 6

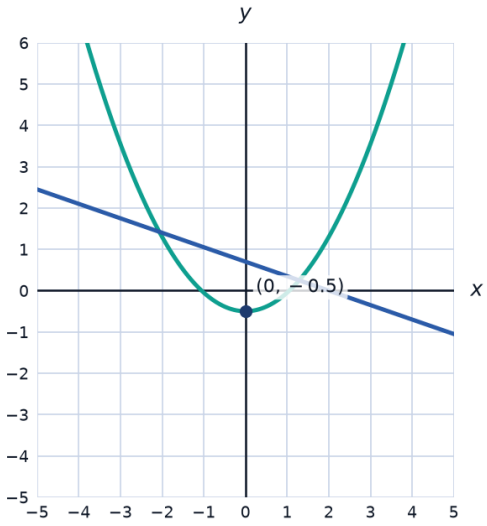
c. 3

d. -3

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33 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as **v**. Find the components of **v**.



a. (6, 4)

b. (10, 10)

c. (-6, -4)

d. (4, 6)

34 [Intermediate] • Target: Intersecting and skew lines

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

a. $x - y + 2z = -4$

b. $x + y + 2z = 6$

c. $x - y + 2z = 4$

d. $x + y = -2$

35 [Intermediate] • Target: The equation of a plane

Line integral of scalar field: $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\,ds =$

a. area

b. volume

c. 0

d. arc length

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36 [Intermediate] • Target: Angles between lines, planes, and pairs of planes

Where does the line $\mathbf{r}(t) = \langle 3, -3, -1\rangle + t\langle 0, 0, 1\rangle$ meet the plane $3x + y + 3z = 4$?

a. $\left(3, -3, \frac{1}{3}\right)$

b. $\left(3, -3, -\frac{2}{3}\right)$

c. $\left(3, -3, -\frac{4}{3}\right)$

d. $(3, -3, -1)$

37 [Intermediate] • Target: Distances and same-side reasoning

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

a. $(3, -12, -2)$

b. $(4, -10, -3)$

c. $(2, -10, 0)$

d. $(5, -14, 0)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a drone's flight path • Cross-Lesson: 10.01 Numerical Methods

Find the angle between gradient of $f = x^2 + y^2 + z^2$ at $(1, 1, 1)$ and gradient of $g = x + 2y + 3z$:

a. 0

b. $\theta = \arccos \frac{6}{\sqrt{12}\sqrt{14}}$

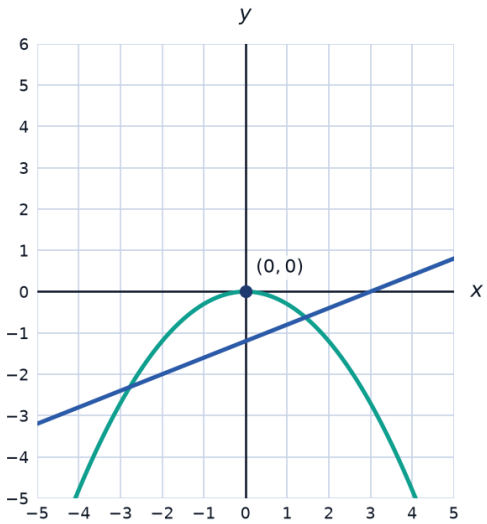
c. $\pi/2$

d. π

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39 [Challenging] • Target: Vector arithmetic in three dimensions • Cross-Lesson: 11.01 Differential Equations

Line 1: $\mathbf{r} = (-2;\ -4; 2) + s(-2; 3;\ -1)$. Line 2: $\mathbf{r} = (0;\ -1; 4) + t(-4; 6;\ -2)$. Are the lines parallel?



a. Yes

b. No

c. They are the same line

d. They intersect at one point

40 [Challenging] • Target: The scalar (dot) product and the angle between vectors • Cross-Lesson: 12.01 Poisson Distribution

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5;\ -2; 2)$ passing through $P(-3,\ -4,\ -3)$.

a. $-5x - 2y + 2z = 20$

b. $-5x - 2y + 2z = 18$

c. $-5x - 2y + 2z = 17$

d. $-5x - 2y + 2z = 14$

41 [Challenging] • Target: The vector equation of a line • Cross-Lesson: 13.01 Continuous Random Variables

Equation of plane through $(1, 0, 0)$ with normal $\mathbf{n} = (1, 2, 3)$:

a. $x + 2y + 3z = 1$

b. $x + 2y + 3z = 0$

c. $2x + y + 3z = 1$

d. $x = 1$

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42 [Challenging] • Target: Classifying lines: parallel, perpendicular and collinear points • Cross-Lesson: 14.01 Sampling & Estimation

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4; -1; 0)$.

a. 90°

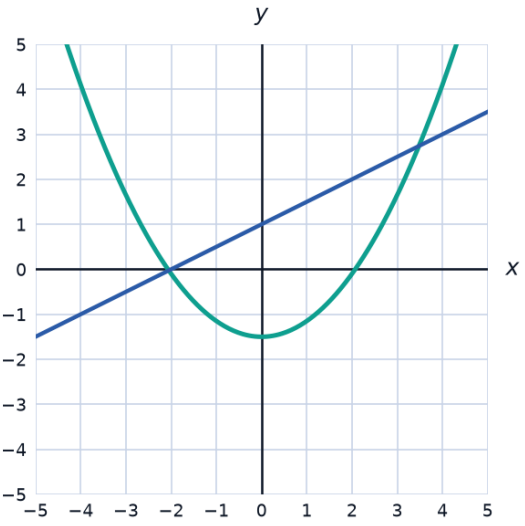
b. 45°

c. 60°

d. 30°

43 [Challenging] • Target: Intersecting and skew lines • Cross-Lesson: 15.01 Hypothesis Testing

Two planes are parallel if their normal vectors are:



a. parallel (scalar multiples)

b. perpendicular

c. equal in magnitude

d. zero

44 [Challenging] • Target: The equation of a plane • Cross-Lesson: 2.01 Exponentials & Logarithms

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. 4/3

c. -3/4

d. -4/3

45 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 3.01 Advanced Trigonometry

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

a. $(3; 4; 6)$

b. $(5; 8; 12)$

c. $(-3; -4; -6)$

d. $(4; 6; 9)$

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